

School Safety for Earthquakes



What To Do Before An Earthquake



- ★ Develop an Earthquake Contingency Plan.
- ★ Craft and disseminate a Student-Family Reunification Plan.
- ★ Create a database of students and their family contact details.
- ★ Train teachers on alternative delivery modes (ADM) of learning.
- ★ Post emergency hotlines in classrooms and in conspicuous areas.
- ★ Coordinate with the LGU on the locations of evacuation areas outside the school, and safe areas where learners & their families can go.
- ★ Advise learners and their families to secure their family disaster kits at home.
- ★ Store vital school documents, learning and teaching materials, computers and equipment in safe buildings and storage place.
- ★ Keep flammable or hazardous liquids in a secure location.
- ★ Conduct and participate in regular earthquake and fire drills (DO 48 s. 2012) with local DRRM councils and other agencies.
- ★ Establish evacuation protocols, including evacuation procedures for elderly, young, children, and persons with disabilities.
- ★ Equip the school with adequate first aid kits, fire extinguishers, and supplies needed for emergencies (e.g. flashlight, megaphone, generator).
- ★ Equip learners and personnel with a GO-BAG.

Anatomy of an Earthquake

Epicenter

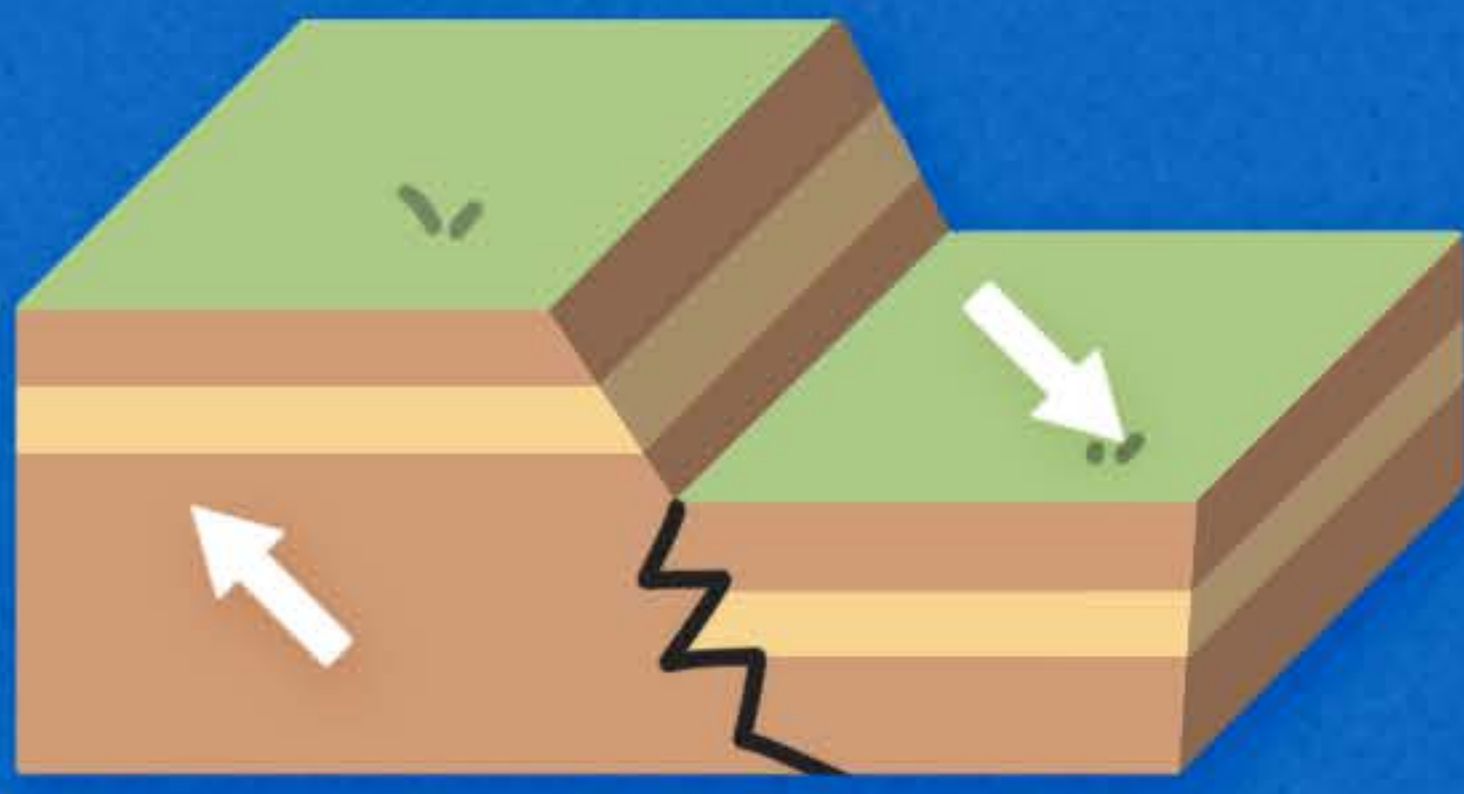
(point on the surface
directly above the focus)



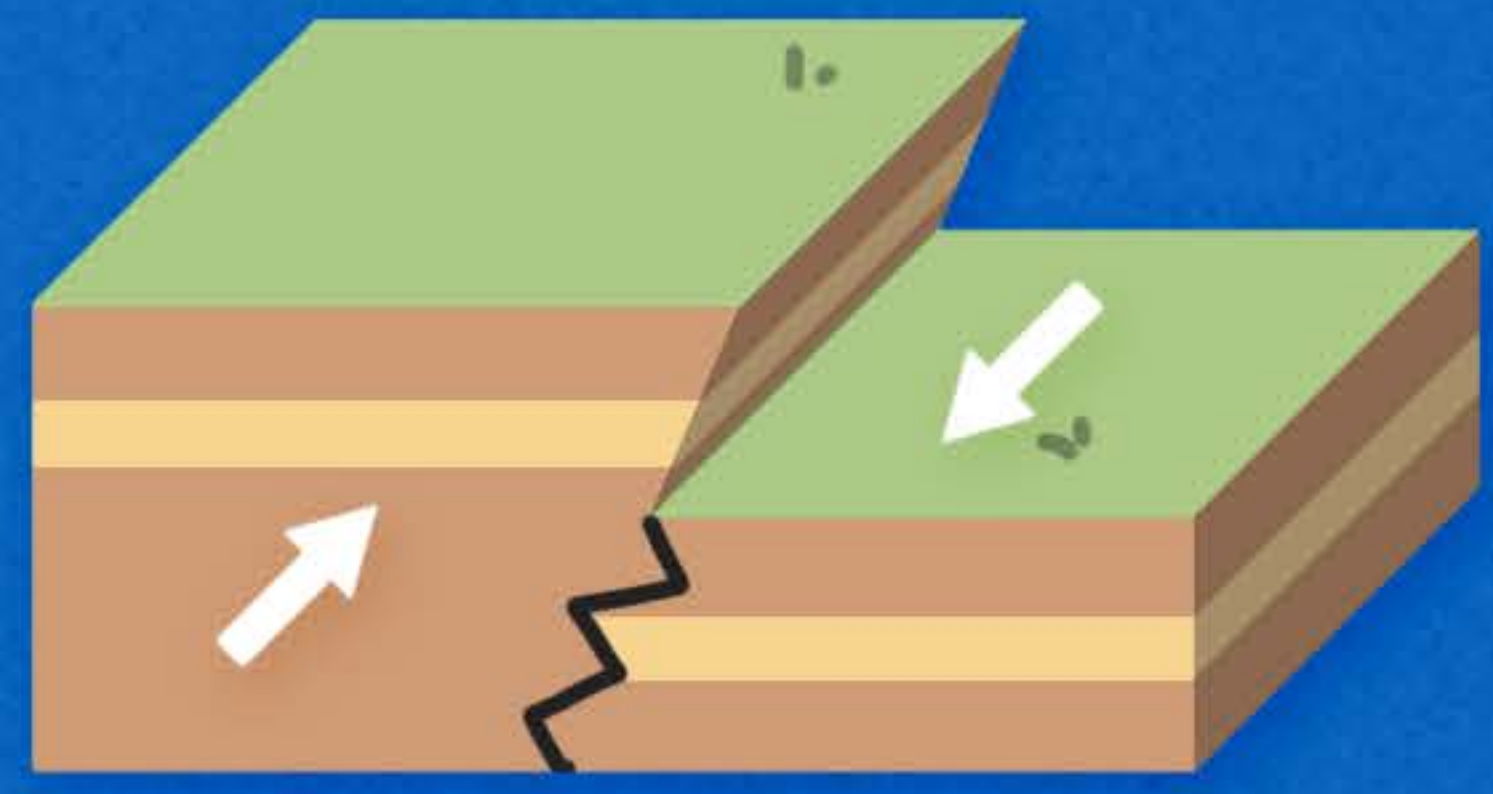
Focus

(center of
energy
released)

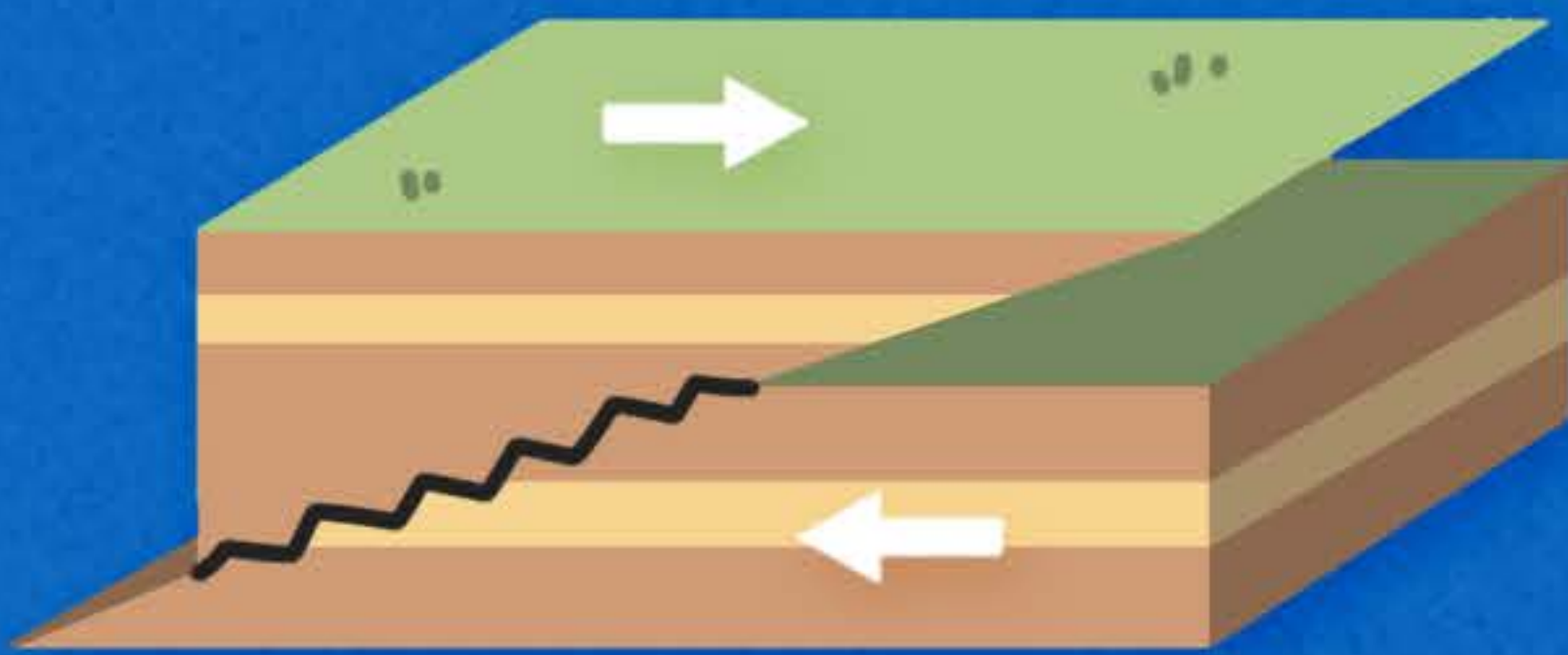
Types of Faults



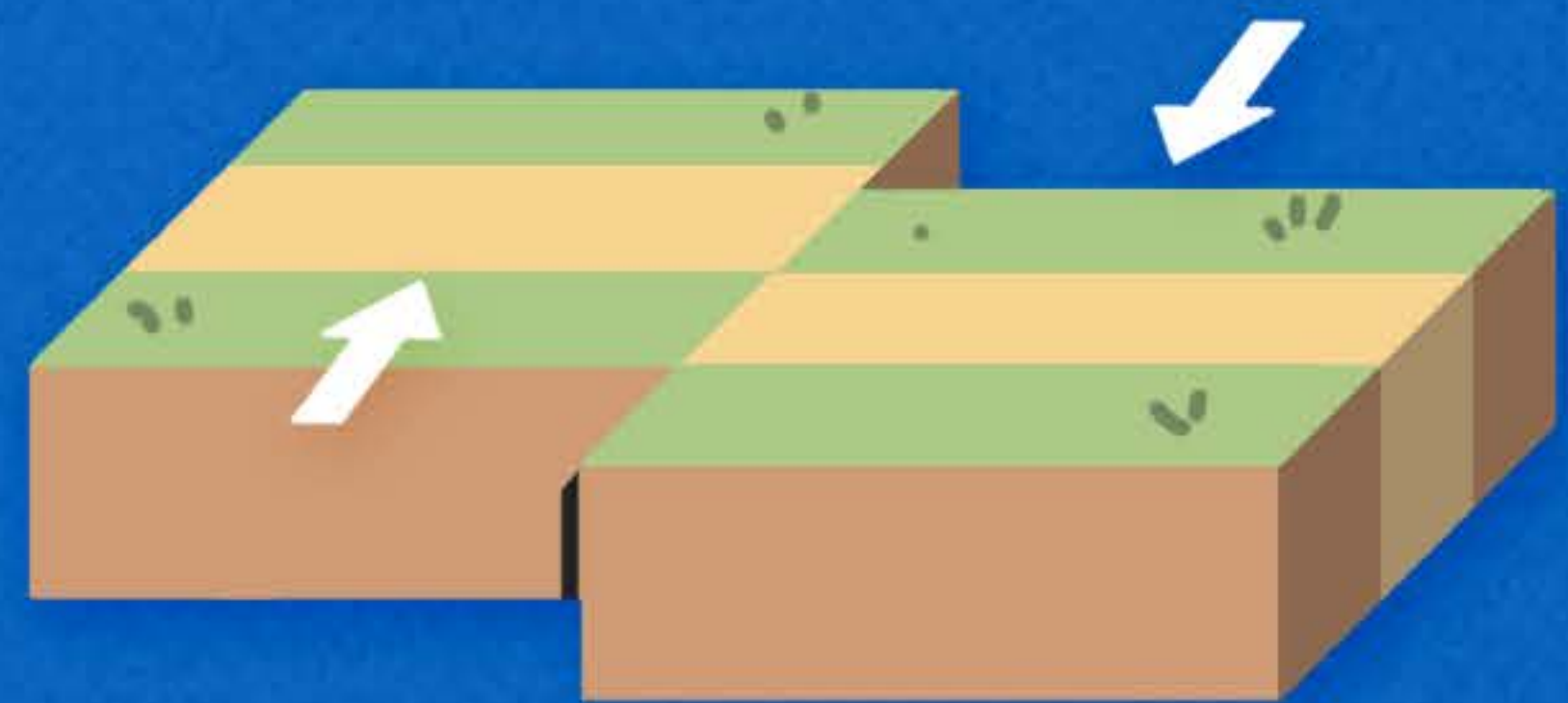
Normal



Reverse



Thrust



Strike-Slip

Intensity

(effect or
felt strength
at the surface in
a certain location)

VS

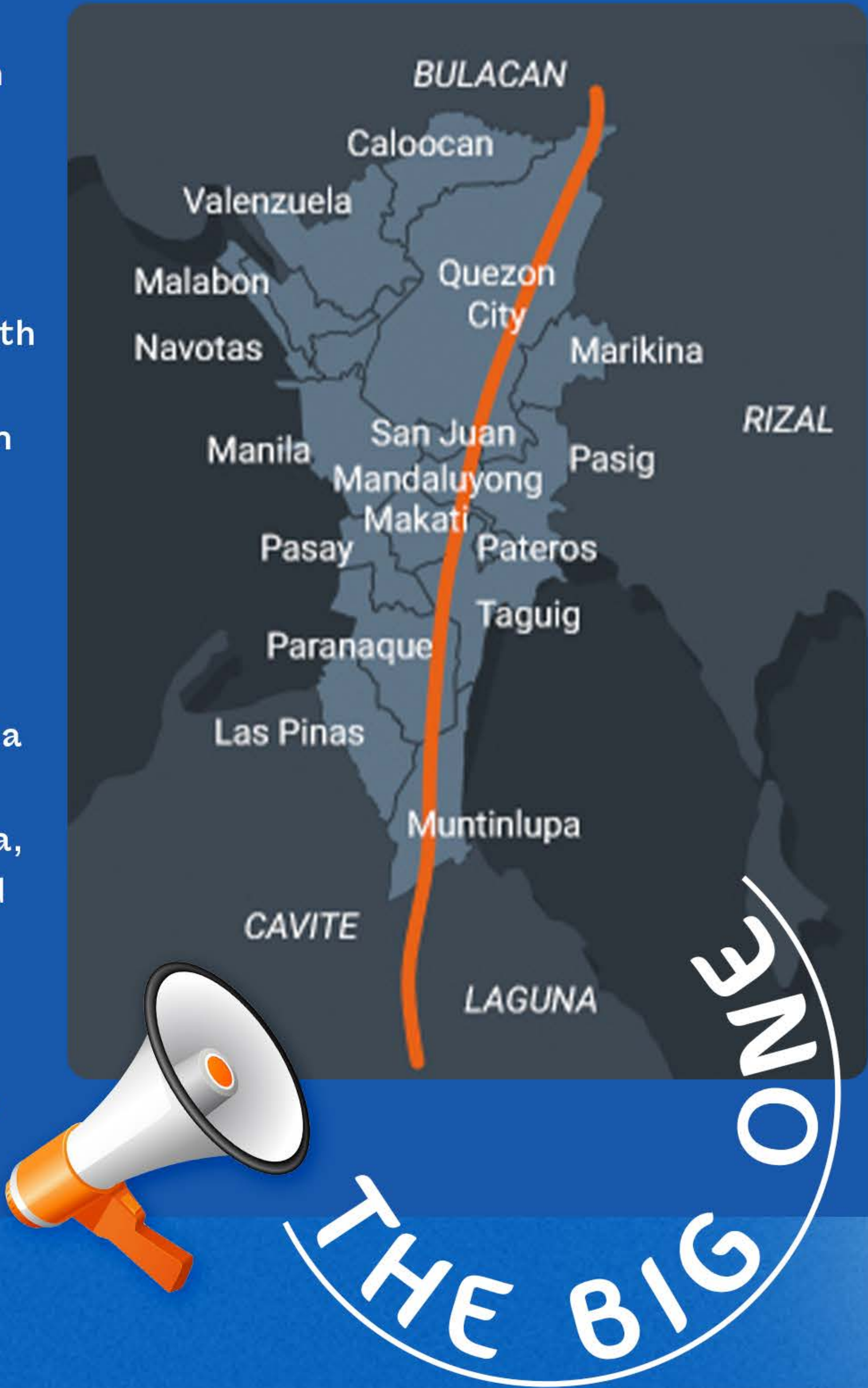
Magnitude

(energy released
from the focus)



The West Valley Fault (WVF) has moved four (4) times within the last 1400 years. The approximate return period of these earthquakes is less than 500 years and no event along the WVF is known after the 17th century. Specifically, the last earthquake that occurred was in 1658, meaning that the anticipated movement of the WVF is approaching.

The WVF transects what is called the Greater Metro Manila Area which consists of Metro Manila, Bulacan, Cavite, Laguna, and Rizal. The WVF is expected to generate a 7.2 magnitude earthquake. According to PHIVOLCS, this magnitude can yield an intensity 8 earthquake that can affect Metro Manila and nearby regions of Central Luzon and CALABARZON.

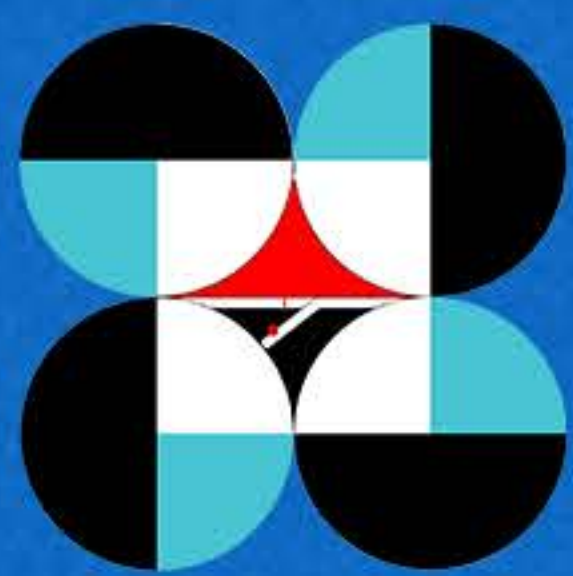


Exposed Public & Private Schools and Learners S.Y. 2017-2018

(taken from the Enhanced Basic Education Information System)

REGION	PUBLIC SCHOOLS	Pub. School LEARNERS	PRIVATE SCHOOLS	Priv. School LEARNERS
NCR	792	1,932,629	2,412	671,498
Region III	669	591,494	620	203,325
CALABARZON	1,397	1,601,781	2,137	615,716

★ Given the lack of adequate open spaces, the schools, in coordination with their Local Government Units (LGUs), need to identify and designate safe areas for evacuation of learners and personnel.



PHIVOLCS Earthquake Intensity Scale

I - Scarcely Perceptible

II - Slightly Felt

III - Weak

IV - Moderately Strong

V - Strong

VI - Very Strong

VII - Destructive

VIII - Very Destructive

IX - Devastating

X - Completely Devastating

What Happens During an Earthquake

You will see...

hanging objects like lamps, wall frames, etc. swing violently or fall down. Books, cabinets, tables, and chairs move or fall over. Window panes rattle or break.

You will feel...

a weak or very strong shaking during an earthquake. The shaking may start out gently and then it becomes strong enough to knock a person off their feet.

You will hear...

low and loud rumbling noises, followed by sounds of shaking, crackling and creaking wood, breaking glass and other falling objects.



Common Hazards may include...



large & unsecured pieces of furniture that could topple



obstructions that may block exit points and corridors



hanging objects like picture frames or flower pots



pieces of glass that could fall or break

When the shaking starts...



... stay put until the shaking stops.

After an Earthquake



Do...

1. Get out of the classroom in an orderly manner

2. Bring your GO BAG with you

3. Proceed to identified evacuation area



Do not...

Panic:
x. Run
x. Push
x. Talk

Use any elevator

Go near buildings, tall trees, powerlines

- ★ If you are located in a tsunami-prone area, immediately head to a higher ground.
- ★ Be careful of falling debris, broken glasses, and other hazardous materials.

Sources:

- ★ "Earthquake Safety in Schools: A Primer For Teachers". DOST-PHIVOLCS. www.phivolcs.dost.gov.ph/images/IEC/eq_safety_in_school.pdf. Retrieved February 19, 2019.
- ★ "PHIVOLCS Earthquake Intensity Scale (PEIS)". DOST-PHIVOLCS. www.phivolcs.dost.gov.ph/index.php?option=com_content&task=view&aid=456&tmid=100. Retrieved February 19, 2019.
- ★ "Metropolitan Manila Earthquake Impact Reduction Study (MMEIRS)". JICA, MMDA & DOST-PHIVOLCS. ndrrmc.gov.ph/attachments/article/1472/Earthquake_Impact_Reduction_Study_Volume_1.PDF. Retrieved February 15, 2019.
- ★ "Living Safely With Nature's Hazards: The Tole of PHIVOLCS In Disaster Risk Reduction." Perez, Jeffery (2016)

For teachers/school personnel...

1. Check yourself and others for injuries. Provide first aid if able.

2. Do a headcount of students and personnel.

3. If there are missing/trapped students and personnel, call for emergency assistance from local authority to conduct rescue operations.

4. Check water and electrical lines for damages.

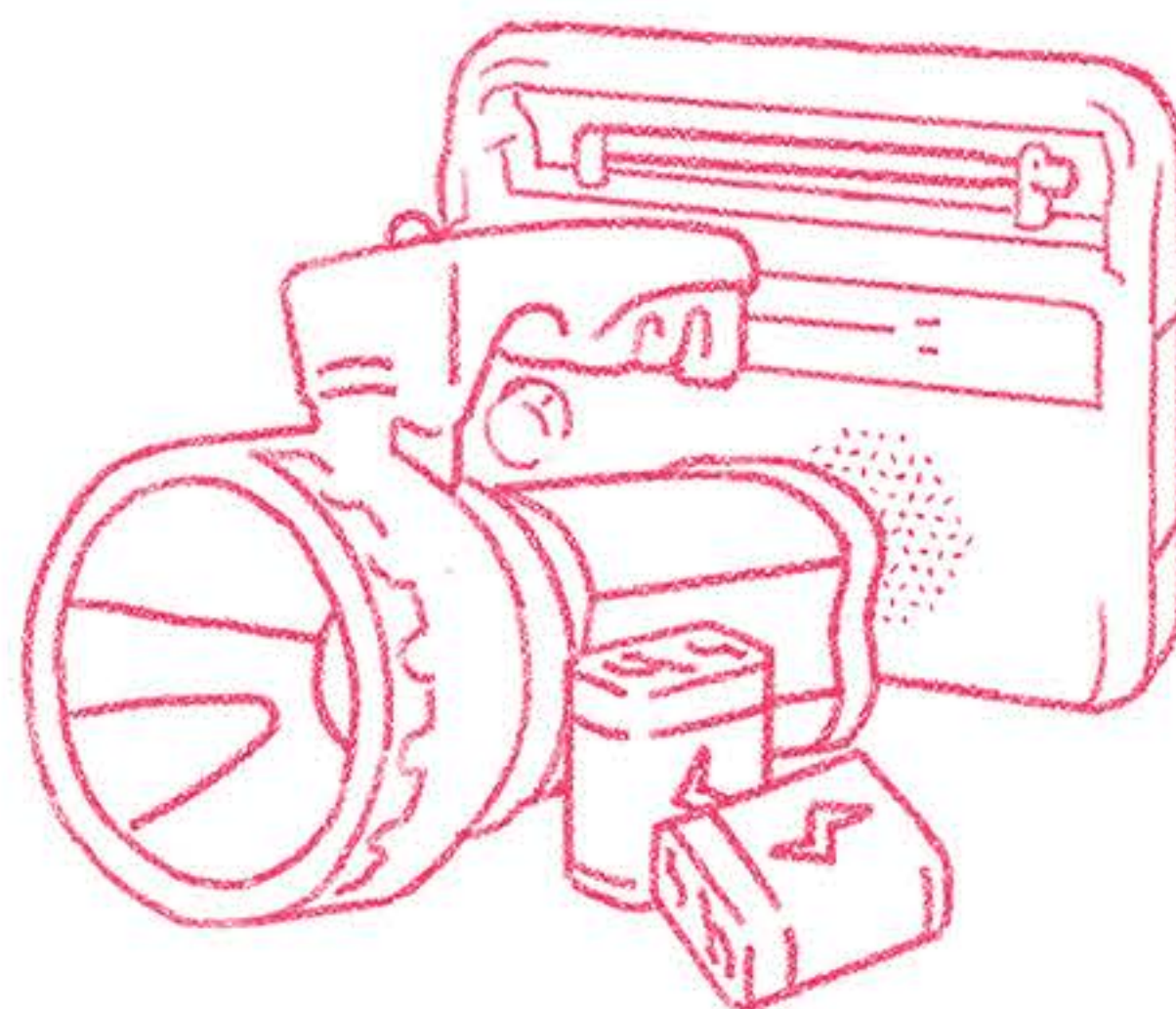
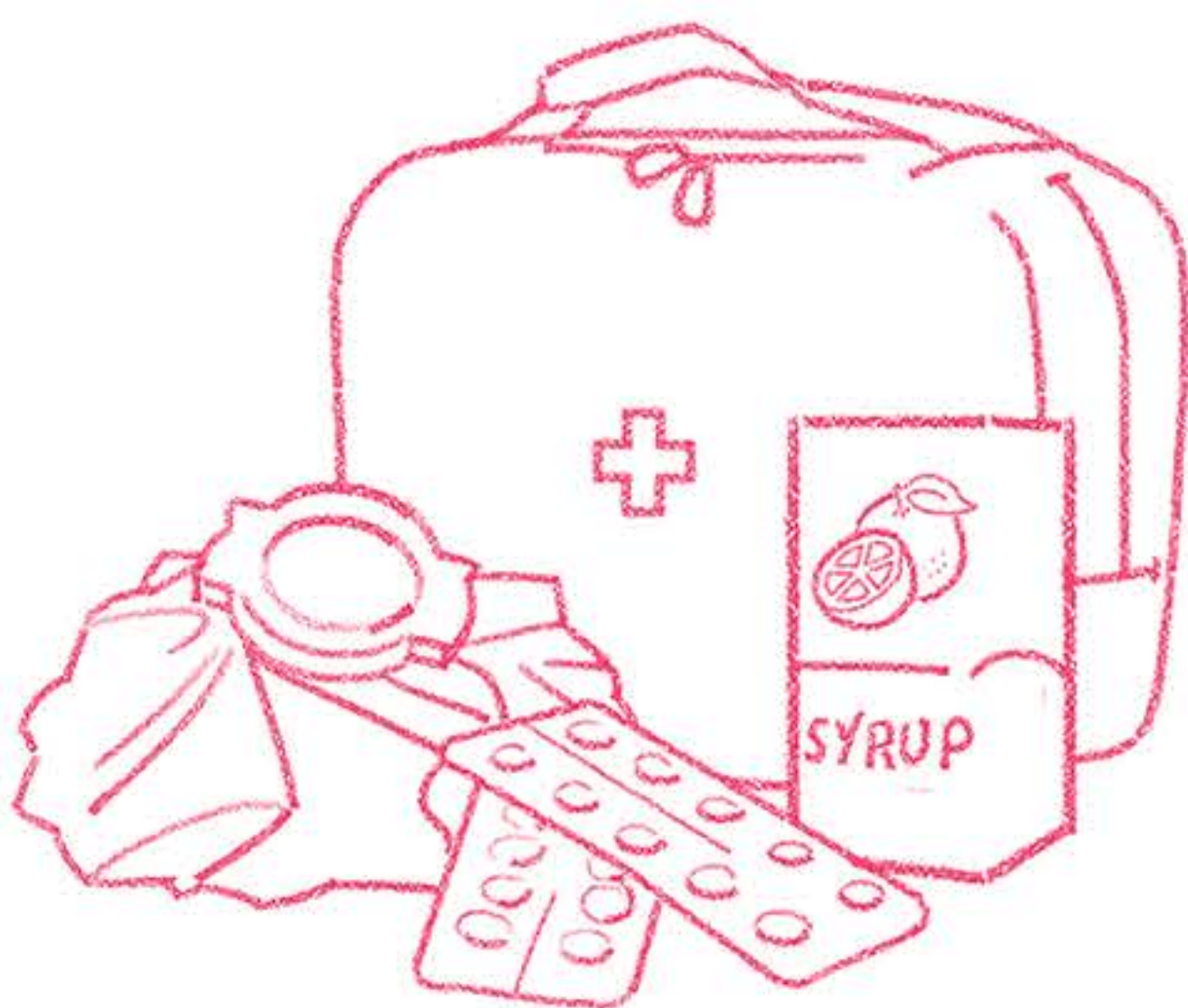
5. Check for spills of chemical, toxic, and flammable materials.

6. Control fires which may spread.

7. Be alert and prepared for secondary hazards caused by earthquake: fire, landslide, and tsunami.

8. Do not enter/re-enter buildings unless cleared.

9. Follow the school contingency plan for earthquakes.



72-hour Family Disaster Kit

★ FIRST AID KIT

with PRESCRIPTION
MEDICATIONS

★ WATER

[2.5L per PERSON, per DAY]

★ WATER

PURIFICATION TABLETS

★ NON-PERISHABLE FOOD

★ DUST MASK

★ WHISTLE

★ IDs / IMPORTANT DOCs

★ EMERGENCY

CONTACT NUMBERS

★ EXTRA MONEY

★ PORTABLE RADIO

★ CELLPHONE

★ FLASHLIGHT

★ EXTRA BATTERIES

★ MATCHES/LIGHTER

★ CANDLES

★ ROPE

★ MULTIPURPOSE KNIFE

★ TOILETRIES

★ RAIN COAT & BOOTS

★ DRY CLOTHES

★ WARM BLANKET

★ SLEEPING BAG/MATS

★ PLATES & UTENSILS

★ GARBAGE BAG

★ SPECIAL NEEDS

OF ELDERLY, WOMEN,

CHILDREN, INFANTS,

PERSONS W/ DISABILITY



Keep this list of hotlines and social media accounts handy during emergencies:

National Disaster Risk Reduction and Management Council (NDRRMC)

TELEPHONE HOTLINES: (02) 911-1406, (02) 912-2665, (02) 912-5668, (02) 911-1873

FACEBOOK: **NDRRMC**

TWITTER: **@NDRRMC_OpCen**

REGIONAL HOTLINES IN LUZON:

NATIONAL CAPITAL REGION: (02) 421-1918

CENTRAL LUZON: (045) 455-1145

CALABARZON: (049) 531-7266

Office of Civil Defense (OCD)

FACEBOOK: **Civil Defense PH**

TWITTER: **@civildefensePH**

REGIONAL HOTLINES:

NATIONAL CAPITAL REGION: (02) 913-2786

CENTRAL LUZON: (045) 455-1526

CALABARZON: (049) 834-4244, 531-7279

Philippine National Police (PNP)

HOTLINE: 117

SMS: SEND TXT PNP TO 2920

FACEBOOK: **Philippine National Police**

TWITTER: **@pnppio**

Bureau of Fire Protection (NCR)

HOTLINE: 117, (02) 729-5166, (02) 410-6319

Department of Social Welfare and Development (DSWD)

HOTLINE: (02) 951-7119

DISASTER RESPONSE UNIT:

(632) 931-81-01 TO 07, LOCAL 426

EARTHQUAKE TEXT HOTLINES:

FOR GLOBE AND TOUCH MOBILE USERS, TEXT "IREPORT<SPACE>NAME/LOCATION/MESSAGE" TO 2327 OR 0917-890-2327.

FOR SMART, SUN AND TALK 'N TEXT USERS, TEXT CONCERNS TO 0918-912-2813.

FACEBOOK: **DSWD**

TWITTER: **@dswdserves**

Department of Transportation (DOTr)

HOTLINES: 7890 OR

(632) 790-8300 AND (632) 790-8400

FACEBOOK: **DOTrPH**

TWITTER: **@DOTrPH**

Metro Manila Development Authority (MMDA)

HOTLINES: 136

TRUNKLINE: 882-4151 TO 77

FACEBOOK: **MMDA**

TWITTER: **@MMDA**

Department of Public Works and Highways (DPWH)

HOTLINES: (02) 304-3713, (02) 304-3904

TWITTER: **@DPWHph**

Philippine Red Cross (PRC)

HOTLINES: 143, (02) 527-0000,

(02) 527-8385 TO 95

TWITTER: **@philredcross**

North Luzon Expressway (NLEX)

HOTLINES: (02) 3-5000 AND (02) 580-8910

TWITTER: **@NLEXtraffic**

Subic-Clark-Tarlac Expressway (SCTEX)

HOTLINE: (045) 459-0522

MOBILE NUMBER: 0920-96-SCTEX (72839)

Skyway System

HOTLINE: (02) 776-7777 (PLDT)

MOBILE NUMBERS: 0917-539-8762 (GLOBE), 0999-888-0893 (SMART), 0932-854-6980 (SUN)

TWITTER: **@SkywaySOMCO**

South Luzon Expressway (SLEX)

HOTLINES: (049) 508-7509, (02) 584-4389

MOBILE NUMBER: 0917-6877539 (GLOBE)